MIL-E-44231A
14 March 1988
SUPERSEDING
MIL-E-44231
30 June 1986

MILITARY SPECIFICATION

ESCALLOPED POTATOES, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers escalloped potatoes thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification for

STANDARDS

MILITARY

* MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

U.S. Standards for Grades of Butter

* (Application for copies should be addressed to the Chief, Dairy Standardization Section, Dairy Division, Agricultural Marketing Service, Room 2750, South Building, P.O. Box 96456, U.S. Department of Agriculture, Washington, DC 20090-6456.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Chairman, Condition of Food Container Committee, Agricultural Marketing Service, Room 0610, South Building, P.O. Box 96456, U.S. Department of Agriculture, Washington, DC 20090-6456.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Application for copies should be addressed to the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

* Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 1111 North 19th Street, Suite 210, Arlington, VA 22209.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

* (Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, Washington, DC 20418.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4, 6.1, and 6.3).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Potatoes, fresh. Potatoes shall be fresh, firm and of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis.

3.2.2 Water. Water used for ice making, formulation, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Shortening, vegetable, powdered. The powdered vegetable shortening shall consist of components which, when combined, will serve as a product whitening agent. The powdered shortening, when used as an ingredient in this product, shall withstand the conditions of thermoprocessing specified in this document without browning. A typical analysis of this powdered shortening follows:

Fat	-	75.0 percent ± 1.5 percent
Protein	-	5.5 percent ± 0.5 percent
Carbohydrates	-	15.0 percent ± 1.0 percent
Moisture	-	2.0 percent maximum

3.2.4 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized modified maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

3.2.5 Butter, lightly salted. The lightly salted butter shall be U.S. Grade A or better of the U.S. Standards for Grades of Butter.

3.2.6 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents.

3.2.7 Broth, chicken. The chicken broth may be purchased frozen or canned (thermostabilized). The broth shall be clear, essentially fat free, with percent solids sufficient to comply with the solids requirements in the sauce formulation. The chicken broth shall have a characteristic mild chicken broth flavor and odor and may contain flavor enhancers approved by the Food and Drug Administration. The chicken broth shall be produced in a USDA inspected plant in accordance with USDA regulations and when frozen shall have been held at 0°F or below for not more than 75 days prior to sauce formulation.

3.2.8 Microcrystalline cellulose. Microcrystalline cellulose shall be a gel forming cellulose compound of 85 parts microcrystalline cellulose and 15 parts sodium carboxymethylcellulose. The microcrystalline cellulose shall be a white water dispersible powder which is of such size that 99.9 percent shall pass through a U.S. Standard No. 60 sieve, 50 percent shall pass through a U.S. Standard No. 325 sieve and shall have an average particle size of 28 microns in dispersion. The microcrystalline cellulose shall have an ash content of not more than 5 percent and a heavy metal content of not more than 10 parts per million. (The pH of a 2-percent dispersion shall be not less than 6 nor more than 8 and shall have an acid demand requiring 2.2 mL of 0.1N hydrochloric acid to change the pH of 100 mL of a 1-percent solids dispersion to a pH of 4). The microcrystalline cellulose shall be insoluble in water and organic solvents, but shall be dispersible in water to form colloidal sols and gels.

3.2.9 Monosodium glutamate (MSG). MSG shall comply with the Food Chemicals Codex.

3.2.10 Onion flavoring. Onion flavoring shall possess a characteristic onion odor and flavor, shall contain little or no fruit sugar or dextrose, and shall be carried on noniodized salt with or without anticaking agents.

3.2.11 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.12 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.13 Pepper, white, ground. Ground white pepper shall be derived from the mature berries of Piper nigrum L., from which the outer covering or outer and inner coverings have been removed. The ground pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. Volatile oil content shall be not less than 1.0 mL/100 g and the white pepper shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

3.2.14 Celery flavoring, soluble. Soluble celery flavoring shall be prepared by properly blending and dispersing a minimum of 4 percent by weight of total dried fruit extractives of Apium graveolens L. on a soluble, dry, edible carrier. The flavoring shall possess a clean, celery-like aroma with a slightly bitter taste, and shall be yellowish grey in color, free flowing, and free from caking and all extraneous matter.

3.3 Preparation and processing. Processing shall be on a continuous basis.

* 3.3.1 Potato preparation. The potatoes shall be cleaned to remove dirt and extraneous material, peeled, washed thoroughly, trimmed, and cut into approximately 1/4 ± 1/16 inch slices. The potato slices shall be blanched sufficiently to prevent discoloration and to remove excess air. The blanched potatoes shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled drained potatoes shall be handled in a manner to prevent discoloration and shall be filled into the tray pack cans within 4 hours after blanching.

3.3.2 Sauce preparation. The sauce shall be prepared from the following ingredients:

Ingredients	Percent by weight
Water	86.67
Powdered shortening	4.00
Starch, modified	3.00
Butter	2.00
Salt 1/	1.25
Chicken broth (15 to 18 percent solids) 2/	1.00
Microcrystalline cellulose	0.50
Monosodium glutamate	0.50
Onion flavoring	0.50
Lecithin	0.25
Onion powder	0.25
Pepper, white, ground	0.07
Celery flavoring	0.01

1/ The total amount of salt in the sauce shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6).

2/ Chicken broth with different solids content shall be adjusted to 15 to 18 percent.

* The microcrystalline cellulose shall be combined with 10 percent of the water and mixed in equipment that provides a high degree of shear, such as a homogenizer, a vertical-cutter mixer, or other similar equipment. Planetary mixers, lightening mixers, or other similar equipment shall not be used. A representative sample of the mixture shall be filled into a viscosimeter sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

holding chamber and held for 5 minutes prior to its release into the measuring chamber. Proper dispersion of the microcrystalline cellulose shall be demonstrated by a flow of not less than 15 cm nor more than 16 cm in 1 minute at 70°F on the Bostwick viscosimeter scale. The starch and powdered shortening shall be mixed with sufficient water to obtain a thin, smooth slurry. The remaining ingredients shall be mixed in a steam jacketed kettle and heated to 180° to 190°F. The starch slurry shall be added with vigorous mixing to obtain a smooth, lump-free mixture. The microcrystalline cellulose mixture shall be added and thoroughly blended. The entire contents shall be heated to 185°F with continuous agitation and held for 5 minutes. The volume of the final mixture shall be adjusted with water before tray filling to compensate for evaporation loss during heating. The sauce shall be maintained in the temperature of 150° to 180°F and filled into tray pack cans within 4 hours after preparation.

* 3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with layered potato slices and sauce such as to conform to the finished product requirements and to the following requirements.

a. Each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to assure compliance with the finished product requirements (see 3.6.n).

b. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

* 3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6 has been achieved. The filled and sealed thermoprocessed tray pack cans shall show no evidence of can swelling when tested for commercial sterility as specified in 4.5.3.4.

* 3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect, insect parts, hair, wood, paper, glass, or metal.
b. There shall be no foreign odor or flavor such as, but not limited to, stale, sour, rancid, musty, or moldy.
c. There shall be no color foreign to the product.
d. Potato texture shall be firm and not mushy, hard, or fibrous.
e. There shall be not less than 85 percent, by weight, whole intact potato slices.
f. The sauce shall be smooth, opaque, moderately thick, and creamy off-white in color.
g. The product shall show no evidence of excessive heating (materially darkened, scorched).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

h. The diameter of the potato slices shall be governed by the diameter of the potatoes from which they have been derived, but at least 50 percent, by weight, of the slices shall be in the range of 1.5 to 2.5 inches in diameter.
i. The average drained weight of potato slices shall be not less than 66.0 ounces.
j. No individual can shall contain less than 64.0 ounces drained weight of potato slices.
k. The average net weight shall be not less than 106 ounces.
l. No individual can shall contain less than 104 ounces of product.
m. The salt content of any individual tray pack shall be not less than 0.5 percent nor greater than 1.3 percent.
n. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of the Title 21, Code of Federal Regulations, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding of Human Foods", and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this document. The contractor shall assure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection necessary to assure compliance with the requirements of this document.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

* 4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, or compliance be verified by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The tray pack cans shall be selected at random from the lot. The tray pack cans shall be heated at least 35 minutes in boiling water, opened, and inspected for the defects listed in table II.

TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	--	0	2
	8	16	1	2
3201 to 35000	13	--	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/

Category	Defect
Major	
101	Texture of potato slices hard, fibrous, or mushy.
102	Sauce consistency not smooth, opaque, and moderately thick.
103	Less than 85 percent, by weight, whole intact potato slices.
104	Sauce not creamy off-white in color.
105	Drained weight of potato slices in a can is less than 64.0 ounces. 3/ 4/
106	Product shows evidence of excessive heating (materially darkened, scorched).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

TABLE II. Product defects 1/ 2/ (cont'd)

Category	Defect
Major	
107	More than 50 percent, by weight, of potato slices measuring less than 1.5 inches or more than 2.5 inches in diameter.

1/ The presence of foreign material (e.g. glass, dirt, insect parts, insects, paper, hair, wood, metal), foreign odor or flavor (e.g. moldy, rancid, musty, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a U.S. Standard 1/4 inch sieve in a manner that will distribute the product over the sieve without breaking the potato slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve and contents shall be immersed in 140° to 160°F water, and the sieve agitated with a vigorous swirling motion for 1 minute taking care to separate the product into individual slices. The sieve and contents shall be removed from the water and then reimmersed momentarily twice in succession. The sieve shall be tilted at approximately a 45° angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4/ The lot shall be rejected if the average drained weight is less than 66.0 ounces.

4.5.3.3 Salt content testing. Nine tray pack cans shall be selected at random from the lot and distributed as follows:

- Three for laboratory analysis
- Three for submission to the contractor
- Three for retention by an AMS agent as reserve samples.

The three tray packs for laboratory analyses shall individually be tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists. Preparation of the samples shall be as follows. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring blender or equivalent. Any test failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

is a major defect. The lot shall be rejected if the salt content of any individual tray pack is less than 0.5 percent or greater than 1.3 percent. Results shall be reported to the nearest 0.1 percent. Analysis of reserve samples at the request of the contractor shall not be permitted unless the original laboratory analysis indicated that the involved lot will be rejected because of noncompliance with the salt requirement. When the reserve samples are analyzed, their analyses for salt shall be considered final. Unused reserve samples shall be returned to the contractor for inclusion in subsequent lots.

* 4.5.3.4 Commercial sterility testing. The sample size shall be one filled, sealed, and thermoprocessed tray pack can selected from each retort batch in the lot. The sample cans shall be tested for sterility by incubating the cans at 95 ± 5°F for 10 days. Any evidence of can swelling shall be considered a critical defect and shall be cause for rejection of the lot.

* 4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be in accordance with 4.5.4.1.

* 4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except that for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, i.e. label raised or peeled back from edges or corners.

* 4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

* 4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and CFR 21, Part 113, Subpart D. Any nonconformance based on observation of can seam teardowns or of record of can seam teardowns is a major defect and shall be cause for rejection of any involved product.

* 4.5.6 Vacuum examination. Cans shall be allowed to cool to 75 ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter straight edge, the same procedure shall be used across the width, in the center of the tray pack can. When examining a ribbed lid, only lay the straight edge

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance is a major defect and shall be cause for rejection of the lot.

* 4.5.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect or illegible.
Arrangement or number of cans not as specified.

4.5.8 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PACKAGING

5.1 Preservation. The product shall be preserved in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 106 ounces of food product, sealed, and thermoprocessed as specified in 3.4 and 3.5. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 90 pounds per base box. The entire inside area of the can shall be coated. Scratches or fractures shall not penetrate through the interior can coating. The interior can coating shall not peel or blister when in contact with the product. The interior can coating shall neither affect nor be affected by the packaged product. A certificate of conformance shall be furnished to certify compliance with the interior can coating requirements. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.1.1 When specified (see 6.1), the base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient temper to protect the product during shipment and storage. The base plate weight of the body and end of each can shall be 98 pounds per base box. The body design shall be reinforced on the bottom with six longitudinal beads or ribs 0.050 $\pm$ 0.010 inches deep. The beads shall be 9 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the bottom. The end design shall be reinforced with eight longitudinal beads 0.018 $\pm$ 0.004 inches deep. The beads

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

shall be 11 inches by 3/4 inch in length and width and shall be equidistant in the center panel of the end. The center four beads on the end design shall be interrupted to provide a 5-inch by 5-inch flat center panel for labeling. The apex of all these beads shall be outward. The interior and exterior can coating requirements shall be as specified in 5.1.1.

5.1.2 Level C. The product shall be preserved as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B, or C as specified (see 6.1).

* 5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with a HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1) except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1 except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

* 5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box and shall be fabricated of the same material as the box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 and 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. Each tray pack shall be labeled with:

- Official establishment number (e.g., est·38) or a three letter code identifying the establishment.
- Lot number 1/
- Product shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as four digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1987 would be coded as 7078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four digit Julian code. Following the four digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

Alternatively, a pre-printed self-adhering 0.002 inch thick clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2.

In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT: Submerge unopened can in boiling water.
Simmer gently for 40 to 45 minutes.
Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

To heat in oven: Either punch several holes in lid of can or open in usual manner leaving the loose lid in place. Place in a 350°F oven for approximately 35 to 40 minutes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, when tray pack cans are fabricated from 90 pounds per base box, the following precautionary markings in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.4, and 6.3).
c. Provisions for approved preproduction samples (see 3.6.1 and 6.3).
d. Level of preservation and packing required (see 5.1 and 5.2).
e. When tray pack can are to be fabricated from 98 pounds per base box (see 5.1.1.1).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Subject term (key word) listing.

Canned foods
Potatoes
Rations
Tray pack

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

6.5 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Preparing activity:

Army - GL

Project No. 8940-0607

Review activities:

Army - MD, TS
Navy - MC
DP - SS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

U.S. ARMY NATICK RESEARCH, DEVELOPMENT
and ENGINEERING CENTER
ATTN: STRNC-ES
Natick, MA 01760-5014

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
U.S. Army Natick Research, Development
and Engineering Center
ATTN: STRNC-ES
Natick, MA 01760-5014

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-44231A	Escalloped Potatoes, Thermostabilized, Tray Pack

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP-1, Feb 86

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T00:48Z
Check the source to verify that this is the current version before use.